

HANS YANG

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Technical Skills

Languages: C, C++, HTML/CSS, Go, Java, JavaScript, Lua Python, R, SQL, TypeScript, VBA

Software & Tech: AWS Lambda, AWS S3, Azure, Docker, Excel/Google Sheets, Git, GitHub, Matplotlib, MongoDB, MySQL, Node.js, NumPy, Pandas, Redis, Vercel, Vite

Projects

🔗 AI Flashcard Study Tool | *AWS Lambda, AWS SAM, Docker, Gemini API, Go, Node.js, Typescript* **2026 – Current**

- Designing a full-stack serverless AI document-processing pipeline using **AWS Lambda**, **AWS SAM**, and **Docker** to transform uploaded files into structured flashcard data through automated text extraction and generation workflows.
- Developing backend services with **Go** and client integrations with **TypeScript/Node.js**, establishing an end-to-end architecture for document ingestion, AI processing, and user-facing content delivery.

🔗 Visulie | *Lua, Roblox Studio, State/Data-Driven Architecture* **2021 – Current**

- Engineered a Roblox Marketplace animation plugin with a custom editor, serialization pipeline, and runtime interpreter for real-time multi-property animation playback, reducing manual scripting effort by up to 90% and generating **150+ sales and \$1.5K+ CAD revenue**.
- **Actively ranked #1 on Google Search**, increasing plugin visibility and outreach through marketplace optimization and community engagement.
- Iterated on a customer-facing product, leveraging user feedback, feature requests, and bug reports to drive continuous improvements and deliver a more robust user experience.

🔗 FPS | *Lua, Roblox Studio* **2023**

- Shipped a simulator-style progression game built around upgrade-driven gameplay loops and progression systems, scaling to **100K+ total visits** and **100+ peak CCU** through iterative gameplay system tuning, economy balancing, and UX-driven front-end design.
- **Featured in the official Roblox Developer 2022 Year-in-Review** for applying Parallel Lua and Canvas Groups with awareness of parallel execution constraints to improve performance and rendering throughput.

🔗 Pocket Rift | *Lua, Roblox Studio, State Machine/Event/Authoritative Server Architecture* **2019 – Current**

- Implemented user tracking through economy and funnel analytics to measure onboarding behavior, enabling iterative optimizations that improved conversion rates by over **185%**.
- Developed a multiplayer networking architecture with client-side prediction, snapshot-based state replication, and delta compression, reducing latency and **decreasing network load by up to 95% over baseline implementations**.
- Engineered reusable engine frameworks including a **Redis-inspired in-memory cache for player session**, and a **React-inspired reactive state management system**, reducing coupling between gameplay systems and enabling scalable client-server architecture.

🔗 SPY Trading Algorithm | *Python, Matplotlib, yfinance* **Jan. 2026**

- Designed and implemented a Python-based systematic trading strategy for SPY, backtested on historical market data with performance analytics, **outperforming buy-and-hold by approximately 54%**.
- Analyzed performance differences between a rule-based trading approach and buy-and-hold, identifying limitations such as sensitivity to market conditions and transaction assumptions.

Work & Volunteer

Freelance/Contract Gameplay Programmer

2022 – Present (Variable Hours/Week)

Systems & Mechanics

Virtual

- Designed and implemented various gameplay systems, such as incremental gathering systems, optimized network handling, scripted visual effects, and modular mob behavioural systems, improving scalability, maintainability, and player experience across multiple game environments.
- Delivered a complete, playable product within strict deadlines by driving end-to-end feature development, collaborating with stakeholders, and rapidly iterating on feedback to achieve a polished player-ready experience.
- Developed and maintained features across multiple live game projects with monetized user bases, creating ongoing revenue generation through reliable systems development and continuous content iteration.